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Fusing Two-Stage LLM Semantic Refinement with Genetic Programming for Robust Alpha Discovery and Degradation Mitigation

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0.5518→0.6365 — Net Sharpe, Nifty 50 (+15.3% relative; peak 0.6465)

9.61% / 10.63% — Annualised net return, deterministic / peak (ceiling 8.41%)

−35%→ −6.8% — OOS Sharpe decay, US S&P 500 (Vanilla GP → ours)

−92%→ −46.7% — OOS Sharpe decay halved on China SSE Composite

38.8 min ▪ $0 — Full local run, 8-bit quantised LLMs, no API cost, T=0.0

1 Why GP Alphas Break: Semantic Overfitting

- **Genetic Programming (GP)** discovers auditable formulaic alphas by searching trees of financial operators — yet it optimises *only* statistical fitness (Rank-IC / ICIR) on historical data.
- GP therefore cannot separate economically grounded relationships from spurious mathematical artifacts, and selects formulas that fit *structural noise*.
- Hall-of-Fame pathologies observed in our runs: `cos(close)`, `sin(price_to_sma20)`, `sub(close, volume)` (a currency price minus a share count), `mul(cos(pe_ratio), ret_5d)`: brilliant in-sample, **decaying −35.03% (US) and −92.08% (China)** out-of-sample.
- Prior LLM remedies are *destructive*: TEDE prunes subtrees syntactically, AlphaAgent rejects candidates terminally. Because > 80% of evolved expressions carry minor dimensional or syntactic imperfections, rejection causes devastating formula attrition and wastes hard-won predictive structure.

Core idea — reject ≠ repair

Instead of discarding a flawed candidate, the refiner **repairs** the flawed sub-expression under a *closed financial grammar* and compiles it back into the evaluation loop, preserving the valid predictive structure that evolution already discovered.

2 Contributions

1. **Dual two-stage framework**: a regime-aware GP ensemble (AFM macro fusion + HMM market states) coupled with a conditional Two-Stage LLM Semantic Refiner in which a semantic failure ($S \leq 2$) triggers domain-constrained repair rather than rejection.
2. **Closed-vocabulary hallucination control**: a strict feature ($\mathcal{F}$) / operator ($\mathcal{O}$) grammar guarantees that **100%** of LLM-repaired formulas compile into valid Abstract Syntax Trees, with zero runtime variable hallucination.
3. **Empirical alpha discovery**: 9.61% deterministic / 10.63% peak annualised net return and net Sharpe 0.6365–0.6465 on a 23-year Nifty 50 panel, beating momentum, low-volatility, mean reversion, Vanilla GP and the pre-LLM regime ensemble.
4. **Cross-market degradation mitigation**: OOS Sharpe decay on the S&P 500 is compressed from −35.03% to −6.77% and halved on the SSE — evidence that closed-vocabulary semantic repair acts as a structural regulariser against historical noise overfitting.

3 Data, Protocol & Evaluation

- **Panel**: NSE Nifty 50 constituents ($N = 50$), T = 5,834 daily bars (2002–2025), feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^{28}$ — price/volume, momentum (5/20/60 d), volatility scalers, drawdowns, RSI, Bollinger bands, valuation ratios, FinBERT news sentiment and macro trend.
- **Objective**: forward 20-day cross-sectional return (Eq. 1); alphas rank stocks cross-sectionally (Eq. 2) and are scored by Spearman Rank IC (Eq. 3) and ICIR (Eq. 4).
- **Strict temporal split**: GP evolution + HMM training 2002–2015 ▪ validation 2016–2018 ▪ out-of-sample test 2019–2025 (1,729 trading days). Annual fundamentals stay fiscal-year aligned and forward-filled to prevent look-ahead leakage.
- **Portfolio**: dollar-neutral top/bottom quintile, 20-day holding horizon, 10 bps round-trip friction. Reported metrics: annualised net return, net Sharpe, Rank IC, ICIR, maximum drawdown, win rate and turnover.

$$r_{i,t+h} = \frac{P_{i,t+h} - P_{i,t}}{P_{i,t}}, \quad h = 20 \text{ d} \qquad (1)$$

$$\mathrm{RankIC}_t = \rho_{\mathrm{Spearman}}\big(\alpha(\mathbf{x}_{i,t}),\ r_{i,t+h}\big) \qquad (3)$$

$$\mathrm{ICIR} = \frac{\overline{\mathrm{RankIC}}}{\sigma_{\mathrm{RankIC}} + \epsilon}, \quad \epsilon = 10^{-8} \qquad (4)$$

4 Theory: Compressing the Hypothesis Space

- **Combinatorial explosion**: with $|\mathcal{O}| = 13$ operators and $|\mathcal{F}| = 28$ terminals, the unrestricted syntactic space at depth D contains $|\mathcal{H}_{\mathrm{raw}}(D)| = |\mathcal{O}|^{2^D-1} \cdot |\mathcal{F}|^{2^D}$ (Eq. 5) — already $\approx 2.4 \times 10^{39}$ distinct trees at depth $D = 4$.
- **Generalisation bound** (Eq. 6): the out-of-sample gap scales with the empirical Rademacher complexity $\mathcal{R}_n(\mathcal{H})$, which follows the pseudo-dimension of the hypothesis class. For $\mathcal{H}_{\mathrm{raw}}$ that dimension is enormous, explaining the severe decay of standard GP.
- **Economic projection** (Eq. 7): the closed vocabulary together with the LLM refiner implements $\Pi_{\mathrm{econ}} : \mathcal{H}_{\mathrm{raw}} \to \mathcal{H}_{\mathrm{sound}}$, eliminating dimensionally absurd operators, periodic functions on price levels and spurious high-frequency noise, so that $\mathcal{R}_n(\mathcal{H}_{\mathrm{sound}}) \ll \mathcal{R}_n(\mathcal{H}_{\mathrm{raw}})$.

$$\sup_{f \in \mathcal{H}} \left| \mathcal{L}_{\mathrm{OOS}}(f) - \mathcal{L}_{\mathrm{IS}}(f) \right| \ \leq\ 2\,\mathcal{R}_n(\mathcal{H}) \ +\ \mathcal{O}\left(\sqrt{\frac{\ln(1/\delta)}{n}}\right) \qquad (6)$$

5 Cross-Market Degradation Mitigation

Market	Architecture	IS Sharpe	OOS Sharpe (decay)
US S&P 500	Vanilla GP	1.393	0.905 (−35.03%)
	HMM Regime-GP	1.228	0.974 (−20.71%)
	Proposed framework	1.153	**1.075 (−6.77%)**
China SSE	Vanilla GP	0.436	0.035 (−92.08%)
	HMM Regime-GP	0.633	0.232 (−63.28%)
	Proposed framework	0.126	**0.067 (−46.68%)**
India Nifty 50	Vanilla GP	0.954	0.5518 (−42.16%)
	HMM Regime-GP	0.814	0.5244 (−35.62%)
	Proposed framework	0.957	**0.6365 (−33.49%)**

Degradation is the real target

On the efficient S&P 500 the framework attains the **highest OOS Sharpe (1.075)** from the **lowest** in-sample Sharpe — decay cut **5×** (US) and **∼2×** (China SSE).

6 Friction Sensitivity of Net Alpha (Table 7)

Round-trip friction	Ann. return	Net Sharpe	Max DD
0 bps (frictionless)	11.84%	0.7420	−18.21%
10 bps (baseline)	**9.61%**	**0.6365**	−19.68%
20 bps	7.38%	0.5098	−21.16%
30 bps (stress test)	5.15%	0.3831	−22.65%

Each extra 5 bps of friction costs ∼1.11% p.a. (20-day rebalancing, ∼14.8% turnover); the **38.4 bps break-even** far exceeds the 4–8 bps cost of liquid NIFTY 50 execution, so the alpha is not a churning artifact.

7 System Architecture

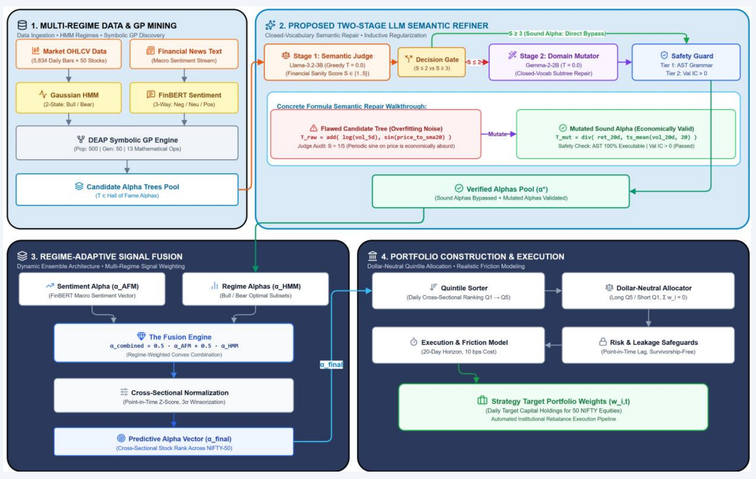


Figure 1. Dual two-stage framework: (1) the regime-aware factor engine (AFM macro regimes + HMM states + regime-specialised GP evolution) and (2) the post-evolution Two-Stage LLM Semantic Refiner (Llama-3B judge, Gemma-2B closed-vocabulary repair at T=0.0) with AST and validation guards.

8 Stage 1 — Regime-Aware Factor Engine

- **AFM macro anchor (FinALFA)**: ∼300k news articles are entity-matched and business-keyword filtered; FinBERT sentiment is fused with sector-standardised fundamental ratios through a learned scalar $\alpha^* \in [0, 1]$:

$$x_t(\alpha) = \alpha^* \tilde{S}_t + (1 - \alpha^*)\tilde{F}_t \qquad (8)$$

- **Gaussian HMM**: a two-state HMM (Baum–Welch, NIFTY-50 index) yields a persistent expansion state (State 0: 72.5% occupancy, $P(0\to0) = 0.994$) and a turbulence state (State 1: 27.5%, $P(1\to1) = 0.983$), giving each GP engine a consistent, regime-delineated training slice.
- **Regime-specialised GP**: two DEAP populations evolve in parallel, each conditioned on its own latent state, with validation ICIR as fitness (population 500, 50 generations, depth ≤ 6, tournament 7, $P_c = 0.7$, $P_m = 0.1$, parsimony $\lambda = 0.002$):

$$\mathcal{P}_{\mathrm{Bull}} \sim \mathrm{Fitness}(\alpha \mid \hat{s}_t = 0), \quad \mathcal{P}_{\mathrm{Bear}} \sim \mathrm{Fitness}(\alpha \mid \hat{s}_t = 1) \qquad (9)$$

- **Economic specialisation**: bull engines select momentum and volatility-scaled terms riding institutional flows; bear engines shift to mean reversion, drawdown and valuation metrics capturing liquidity overreaction.

9 Stage 2 — Two-Stage LLM Refiner

Stage 1 — economic judge (Llama-3.2-3B-Instruct)

Each Hall-of-Fame tree is scored on a **5-point financial sanity rubric** under greedy decoding ($T = 0.0$).

Decision gate: $S \geq 3 \Rightarrow$ **direct bypass**; only $S \leq 2$ is escalated to repair. Neutral formulas (S=3) often encode genuine non-linear structure that heuristics misread — rewriting them destroyed performance.

Stage 2 — domain-constrained mutator (Gemma-2-2B-IT)

Repair is deterministic ($T = 0.0$) and confined *to the closed vocabulary*:

$$\mathcal{T}_{\mathrm{mut}} \sim \mathrm{LLM}_{\mathrm{Gemma}}\big(\mathrm{Prompt}(\mathcal{T}_{\mathrm{failed}}, \mathcal{F}, \mathcal{O})\big) \qquad (10)$$

Two-tier guard: Tier 1 parses the rewrite against $\mathcal{G}(\mathcal{F}, \mathcal{O})$ and checks for variable hallucination; Tier 2 requires $IC_{\mathrm{val}}(\mathcal{T}_{\mathrm{mut}}) > 0$; otherwise the pipeline *safely falls back* to the original candidate.

Final ensemble blend (point-in-time z-score normalisation):

$$\alpha_{\mathrm{final}} = \mathrm{zscore}\big(0.5 \cdot \alpha_{\mathrm{Ensemble,LLM}} \ +\ 0.5 \cdot \alpha_{\mathrm{HMM,LLM}}\big) \qquad (11)$$

10 Closed Vocabulary: Zero-Hallucination Grammar (Table 2)

Features $\mathcal{F}$ (28)	`open`, `high`, `low`, `close`, `volume`, `ret_5d`, `ret_20d`, `ret_60d`, `vol_5d`, `vol_20d`, `vol_60d`, `dd_20d`, `dd_60d`, `price_to_sma20`, `price_to_sma50`, `rsi_14`, `bb_upper`, `bb_lower`, `pe_ratio`, `pb_ratio`, `roe`, `roce`, `debt_to_equity`, `current_ratio`, `sentiment_score`, `afm_fundamental`, `afm_sentiment`, `macro_trend`
Operators $\mathcal{O}$ (13)	`add(a,b)`, `sub(a,b)`, `mul(a,b)`, `div(a,b)`, `neg(a)`, `abs(a)`, `log(a)`, `inv(a)`, `ts_mean(a,d)`, `ts_max(a,d)`, `ts_min(a,d)`, `rank(a)`

100% of repaired expressions compiled to valid ASTs and executed without a single runtime exception across all 1,729 out-of-sample days.

11 Semantic Repair in Action: Three Audited Cases

- **Periodic artifact removal.** Flawed (S=1/5): `add(log(vol_5d), sin(price_to_sma20))`. *Judge:* a sine cannot model price cyclicality — it injects spurious in-sample noise. Repaired (S=4/5): `div(ret_20d, ts_mean(vol_20d,20))`, risk-adjusted momentum; Rank IC −0.012 → +0.038.
- **Dimensional repair.** Flawed (S=1/5): `sub(close, volume)` — a currency price minus a share count. *Judge:* dimensional violation, no economic meaning. Repaired (S=5/5): `(close - ts_mean(close,20))/ts_std(close,20)`, a standardised price z-score.
- **Overfitting-noise smoothing.** Flawed (S=2/5): `mul(cos(pe_ratio), ret_5d)`. *Judge:* cosine modulation of P/E multiples overfits sample noise. Repaired (S=4/5): `div(pe_ratio, ts_mean(pe_ratio,60))` — a relative-valuation discount metric.

12 Compute & Token Economics (Table 8)

- **Where the time goes**: parallel GP evolution dominates at **38.5 min** (0.82 ms/individual); AFM 7.0 s, HMM fit 4.5 s, AST guard 0.02 s — **38.8 min** end-to-end on one local workstation.
- **Refinement is almost free**: only the 50 Hall-of-Fame candidates are refined, so Stage 1 (138 ms/expr, 6.9 s) and Stage 2 (295 ms/expr, 7.4 s) together cost under 15 s; 8-bit weights peak at 4.2 / 3.8 GB VRAM, zero API spend.

13 Reproducible Protocol: Algorithm 1 (Table 3)

$A^* \leftarrow \emptyset$ *(verified alpha pool)*
for each Hall-of-Fame candidate $T \in \mathcal{T}_{\mathrm{raw}}$ **do**
  $S \leftarrow \mathrm{LLM}_{\mathrm{Judge}}(T) \in \{1, \ldots, 5\}$ *Stage 1: Llama-3.2-3B, T=0.0*
  if $S \leq \tau = 2$ **then** $\mathcal{T}_{\mathrm{flawed}} \leftarrow$ flawed sub-expression of T;
    $\mathcal{T}_{\mathrm{mut}} \leftarrow \mathrm{LLM}_{\mathrm{Mutator}}(\mathcal{T}_{\mathrm{flawed}}, \mathcal{F}, \mathcal{O})$ *Stage 2: Gemma-2-2B, T=0.0*
    if AST-ok$(\mathcal{T}_{\mathrm{mut}}, \mathcal{G}(\mathcal{F}, \mathcal{O}))$ **and** $IC_{\mathrm{val}}(\mathcal{T}_{\mathrm{mut}}) > 0$ **then** $A^* \leftarrow A^* \cup \{\mathcal{T}_{\mathrm{mut}}\}$
    else $A^* \leftarrow A^* \cup \{T\}$ *safe fallback*
  else $A^* \leftarrow A^* \cup \{T\}$ *$S \geq 3$: direct bypass (sound alpha)*
return A^* *verified, economically sound alpha set*

14 Main Out-of-Sample Results (Table 4)

Strategy	Ann. net return	Net Sharpe	Max DD
Momentum (12-1M)	−2.57%	−0.1369	−40.65%
Low volatility	−10.24%	−0.5454	−65.71%
Mean reversion	4.46%	0.2981	−20.19%
Vanilla GP	1.62%	0.4417	−4.82%
AFM adaptive fusion	4.08%	0.2785	−19.34%
HMM regime-GP alpha	7.49%	0.5244	−18.17%
Regime-GP ensemble + HMM (pre-LLM)	8.41%	0.5518	−19.62%
Proposed: Llama-3B + Gemma-2B (det.)	**9.61%**	**0.6365**	−19.59%
Proposed: Llama-3.2-3B (peak)	**10.63%**	**0.6465**	−19.59%

Both proposed rows lift the **quantitative ceiling** (8.41% / 0.5518) without inflating drawdown, raising the win rate to **60.4%** and ICIR to **0.1650** (0.1381 pre-LLM). DeepSeek-1.5B and Gemma-2B judges left returns at 8.41% and Qwen-2.5-3B reached 8.60%.

15 Ablation Ladder & Hyperparameters (Table 5)

Configuration	Ann. return	Sharpe	Rank IC	Max DD
Vanilla GP	1.62%	0.44	0.0130	−4.82%
GP (AFM-Bear only)	6.43%	0.48	0.0358	−23.42%
HMM regime-GP alpha	7.49%	0.52	0.0303	−18.17%
Regime-GP soft/hard blend	7.52%	0.56	0.0381	−23.42%
Regime-GP ensemble (pre-LLM)	8.41%	0.55	0.0346	−19.62%
Proposed refiner (full)	**9.61%**	**0.64**	**0.0328**	−19.59%

Monotone ladder: each structural component adds value. Hyperparameter diagnostics (Fig. 5) show maximum tree depth 6 and population size 500 strike the best Sharpe/ICIR trade-off — enough expressive complexity without tree bloat.

16 Two Persistent HMM Market States (Fig. 2)



Figure 2. The two-state HMM separates persistent low-volatility expansion (State 0, green) from high-volatility turbulence (State 1, red), capturing the 2020 COVID collapse and the 2022 tightening correction.

17 Out-of-Sample Wealth Compounding (Fig. 3)

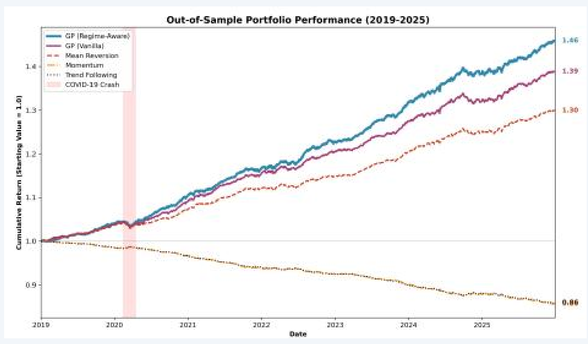
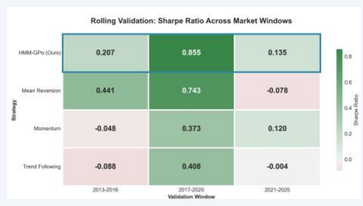


Figure 3. Cumulative net return, 2019–2025: the regime-aware hybrid compounds through bull and bear phases, while momentum and trend following flatten out (−40.65% drawdown for momentum).

18 Rolling Stability & Discovered Features (Figs. 4, 6)



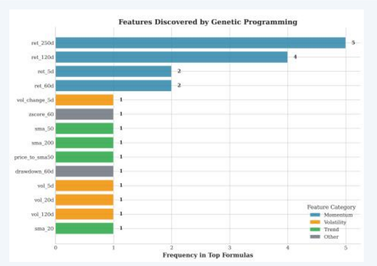


Figure 4 (left). Rolling-window validation Sharpe: HMM-GPα stays positive (0.855) in the turbulent 2017–2020 window where momentum and trend following turn negative. Figure 6 (right). Feature tokens in the top alphas: intermediate momentum and volatility scalers dominate.

19 Conclusions & Outlook

- **Sub-period robustness**: positive net Sharpe in *every* macro sub-period (2019 0.72 ▪ 2020 0.58 ▪ 2021 0.76 ▪ 2022 0.51 ▪ 2023–25 0.74); COVID-shock drawdown −14.82% versus −38.45% for momentum.
- **Economic reading**: the AFM supplies a slow macro anchor while the HMM tracks fast volatility transitions, so evolution finds regime-specific premia instead of averaging contradictory dynamics.
- **Capacity & outlook**: Almgren–Chriss impact implies ≈ $250M deployable before impact erodes 10% of net alpha. Next: LoRA-tuned Gemma-2B, multi-agent consensus judging (Llama-3B + Qwen-3B), scaling to NSE 500 / Russell 2000; current limits are reporting lag and a large-cap-only universe.

Take-away

Closed-vocabulary **semantic repair — not rejection** is a zero-cost structural regulariser that transfers across Indian, US and Chinese equity panels.

Selected references: ▪ Kakushadze, *101 Formulaic Alphas*, Wilmott (2016) ▪ Tang, Sun & Li, *AlphaAgent: automated alpha factor discovery through multimodal LLM agents*, arXiv:2502.16789 (2025) ▪ Chen & Qi, *TEDE: LLM-assisted semantic pruning for GP alpha mining*, Applied Sciences (2026) ▪ Fortin et al., *DEAP: evolutionary algorithms made easy*, JMLR (2012) ▪ Hamilton, *Econometrica* (1989) ▪ Rabiner, *Proc. IEEE* (1989) ▪ McLean & Pontiff, *J. Finance* (2016) ▪ Motghare et al., *FinALFA / multi-document news summarisation*, Computers (2025) ▪ Almgren & Chriss, *J. Risk* (2000).

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